

- | | | | | |
|---|-----|--|--|-----|
| 4 | (a) | + ΔU : increase in internal energy | B1 | |
| | | + q : thermal energy / heat supplied to the system | B1 | |
| | | + w : work done on the system | B1 | [3] |
| | | | | |
| | (b) | (i) | (thermal) energy required to change the state of a substance per unit mass | M1 |
| | | | without any change of temperature | A1 |
| | | | | A1 |
| | | | | [3] |
| | | | | |
| | | (ii) | when evaporating | |
| | | | greater change in separation of atoms/molecules | M1 |
| | | | greater change in volume | M1 |
| | | | identifies each difference correctly with ΔU and w | A1 |
| | | | | [3] |

5 (a) (i) (increased) surface area proportional to

M1